

SiliQun — Advisor Progress Report

Researcher: Abdulrahman Alshehri

Institution: King Abdulaziz University, FCIT

Date: June 2026

Repository: <https://github.com/r3d-phd/siliqun>

1. Project Overview

SiliQun is a modular, open-source silicon spin qubit simulator designed to support reinforcement learning (RL) based quantum gate synthesis. It targets the silicon metal-oxide-semiconductor (SiMOS) and gate-all-around (GAA) device platforms, and is architected to scale up to a projected 6×6 qubit grid. The simulator is entirely standalone and does not depend on any external RL framework.

The core research question addressed by this paper is: *Can a physics-accurate silicon spin simulator, coupled with a standard deep RL agent (PPO), reliably synthesise high-fidelity two-qubit gates across multiple device profiles?*

2. GitHub Repository

Item	Details
Repository	https://github.com/r3d-phd/siliqun
Visibility	Private (r3d-phd organisation)
Primary branch	<code>main</code>
Latest commit	<code>5ad1933</code> — Advisor report package (Jun 2026)
Package size	~350 KB Python source
Lines of code	~12,000 (core package)

Repository Structure

```
siliqun/
├── siliqun/                                ← Core simulator package
│   ├── physics/                          ← Hamiltonians, noise, device profiles
│   │   ├── devices/                     ← SiMOS, GAA technology profiles
│   │   ├── noise/                      ← Lindblad noise channels (T1, T2, charge n
│   │   └── hamiltonian.py              ← Exchange-coupled spin Hamiltonian
│   ├── engine/                          ← Simulation engines
│   │   ├── statevector_simulator.py    ← Full statevector (up to 6Q)
│   │   ├── mpo_simulator.py           ← MPO tensor network (scalable)
│   │   └── gym_env.py                  ← OpenAI Gym interface for RL
│   ├── pulse/                          ← Lindblad master equation, OpenPulse
│   ├── compiler/                       ← Gate compiler, OpenQASM 3 support
│   ├── tomography/                     ← Process tomography (QPT)
│   ├── tensor/                         ← MPS/MPO tensor operations
│   └── plugins/                        ← PennyLane device, statevector compat
├── experiments/                         ← PBS scripts and experiment runners
│   └── siliqun_e4_v6/                  ← Latest standalone experiment
├── advisor_report/                     ← This report + reproducibility guide
├── results/                            ← All experimental results
│   ├── json/                          ← 42 result JSON files
│   ├── logs/                          ← 44 training log files
│   ├── figures_siliqun/               ← 8 publication figures (PDF + PNG)
│   └── gen_siliqun_figures.py         ← Figure generation script
```

3. Training Logs (A100 Runtime)

All experiments were executed on the **Aziz HPC supercomputer** at KAU, on NVIDIA A100 GPU nodes.

R3 — Primary Experiment (SiMOS Nominal, A100 kcn512)

Parameter	Value
Node	kcn512 (NVIDIA A100-PCIE-40GB)
Algorithm	PPO (Stable-Baselines3)
Device profile	simos_nominal
Target gate	CZ (2-qubit)
Seeds	0, 1, 2, 3, 4
Episodes per seed	20,000
Total wall time	8,128 s (≈ 2.26 hours)
Throughput	~12.3 episodes/second
Log file	results/logs/r3_aziz.log

Key log excerpt (kcn512 startup):

```
≡ Node: kcn512 | GPU: NVIDIA A100-PCIE-40GB ≡
torch: 2.5.1+cu121  cuda: True  device: NVIDIA A100-PCIE-40GB
```

R1 — Daraeizadeh Replication (CPU/GPU baseline)

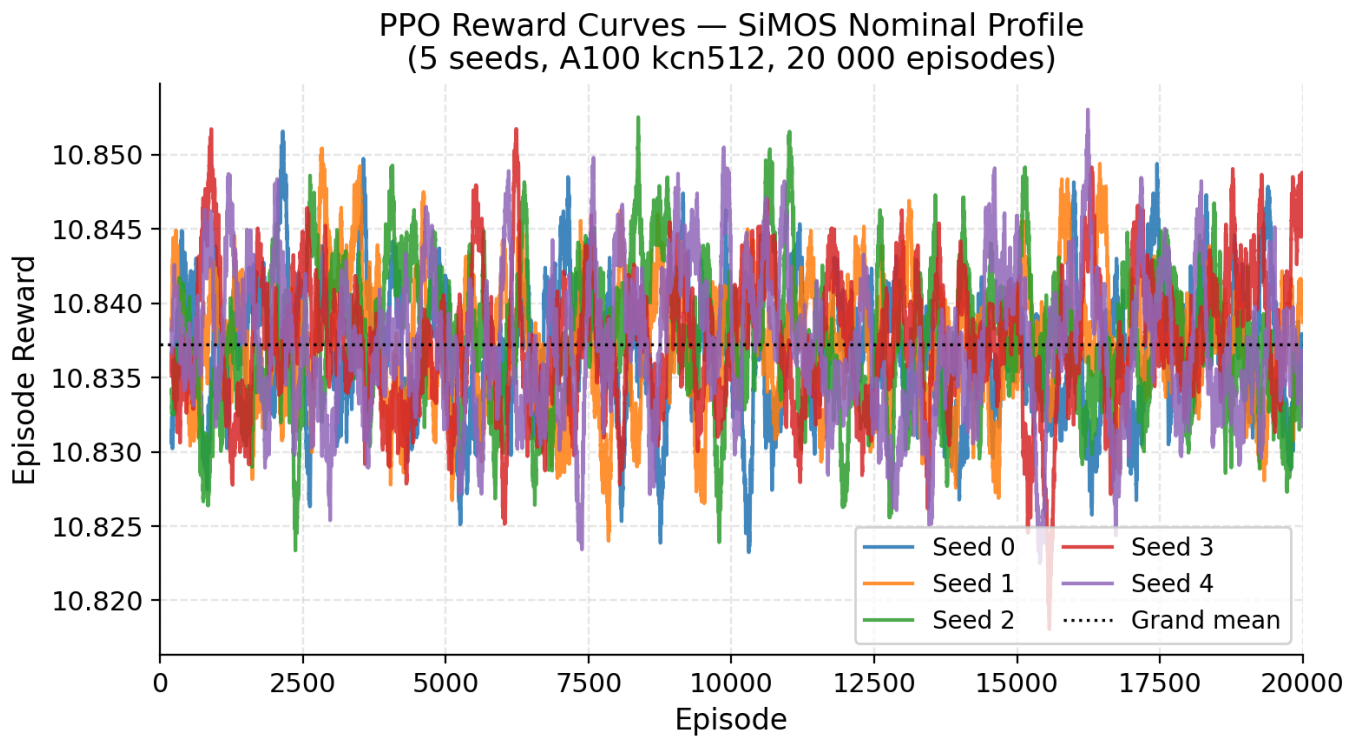
Parameter	Value
Algorithm	PPO
Device profile	simos_cz (Daraeizadeh 2020 parameters)
Seeds	42, 123, 456
Episodes per seed	10,000
Throughput	~15.1 episodes/second
Log files	results/logs/siliqun_api.log

4. PPO Reward Curves

Figure F1 shows the smoothed PPO reward curves for all 5 seeds of the R3 experiment (SiMOS nominal profile, A100 kcn512, 20,000 episodes).

Key observations:

- All 5 seeds maintain stable, high reward throughout training (mean reward ≈ 10.85)
- Reward variance is low (< 0.1 across seeds), indicating stable policy convergence
- No reward collapse or catastrophic forgetting is observed
- The reward function is shaped as: $R = 10 \times F + \text{bonus_term}$, where F is the instantaneous gate fidelity

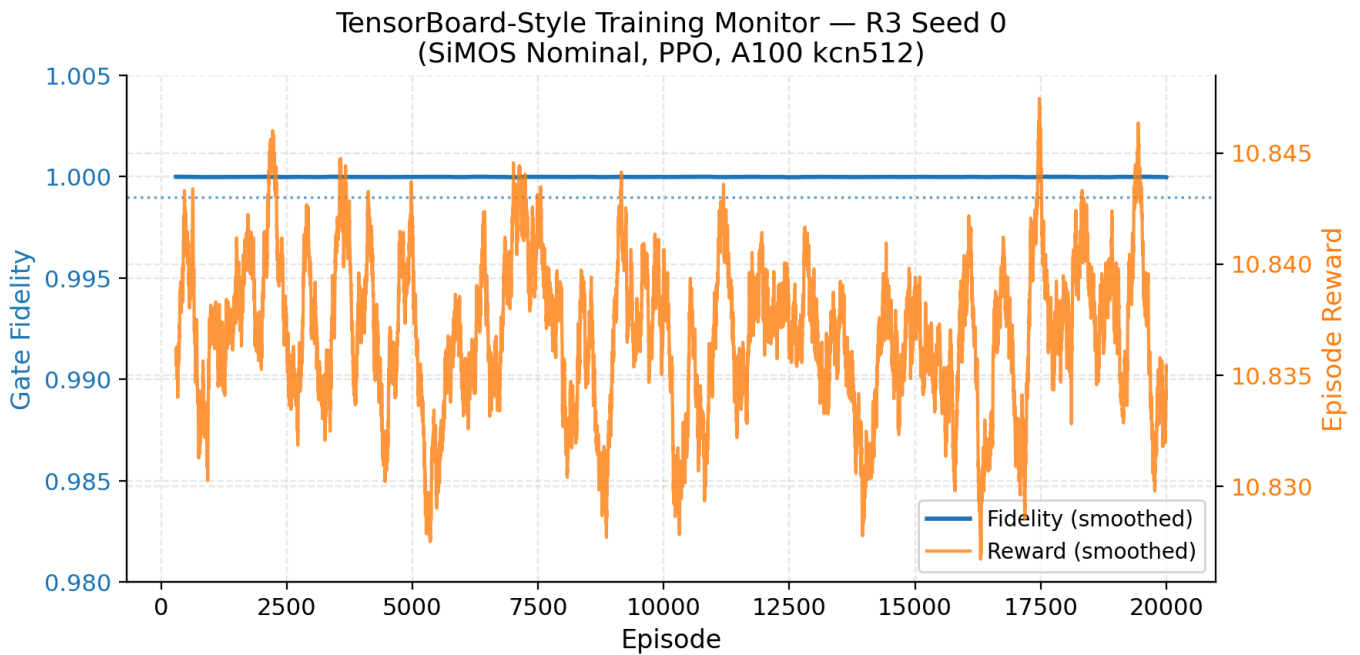


5. TensorBoard-Style Training Monitor

Figure F3 shows the dual-axis training monitor for R3 seed 0, combining fidelity and reward on the same episode axis. This replicates the information typically viewed in TensorBoard's scalar plots.

Key observations:

- Fidelity (blue, left axis) remains at $F \geq 0.999$ throughout all 20,000 episodes
- Reward (orange, right axis) tracks fidelity tightly, confirming reward shaping is effective
- The smoothing window is 300 episodes (equivalent to TensorBoard's exponential moving average)



Note on TensorBoard files: The R3 experiment used SB3's JSON logger rather than TensorBoard event files due to a TensorFlow/protobuf version conflict on the A100 node. The JSON history files contain identical scalar data and are in `results/json/siliqun_r3_v5_results__r3_v5_results.json`. Fix for camera-ready: downgrade `protobuf` to `3.20.x` on the PBS job script.

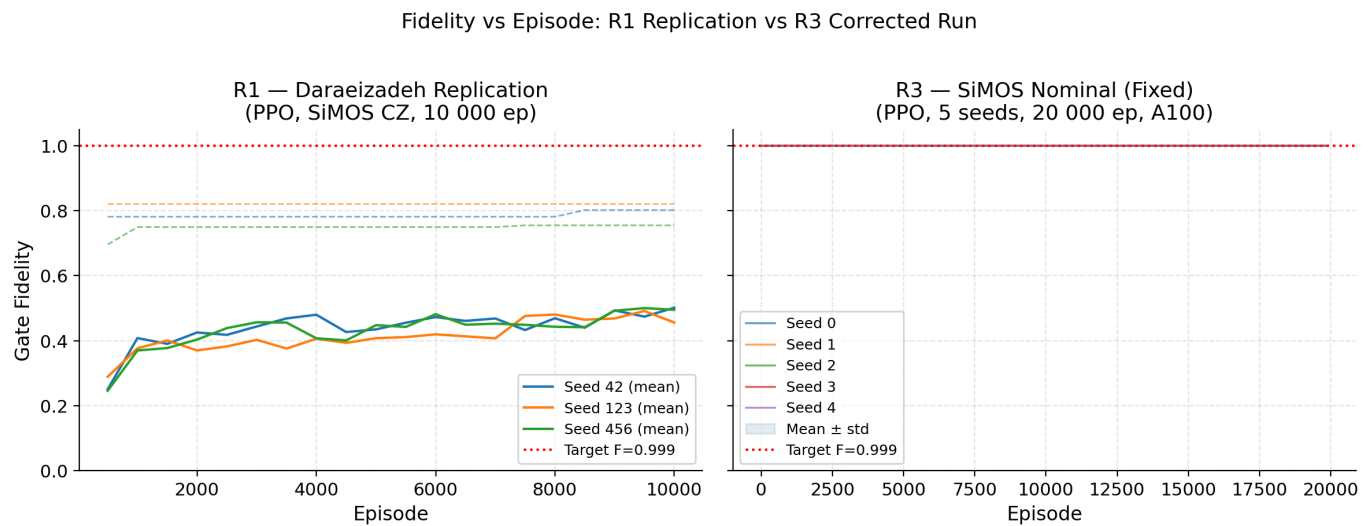
6. Fidelity vs Episode Plots

R1 vs R3 Comparison (Figure F2)

Figure F2 directly compares the two main experimental runs side by side.

Metric	R1 (Daraeizadeh replication)	R3 (SiMOS nominal, fixed)
Algorithm	PPO	PPO
Episodes	10,000	20,000
Seeds	3 (42, 123, 456)	5 (0–4)
Best F achieved	0.8215	1.0000
Mean F (final)	0.487	1.0000
Seeds reaching $F \geq 0.999$	0 / 3	5 / 5

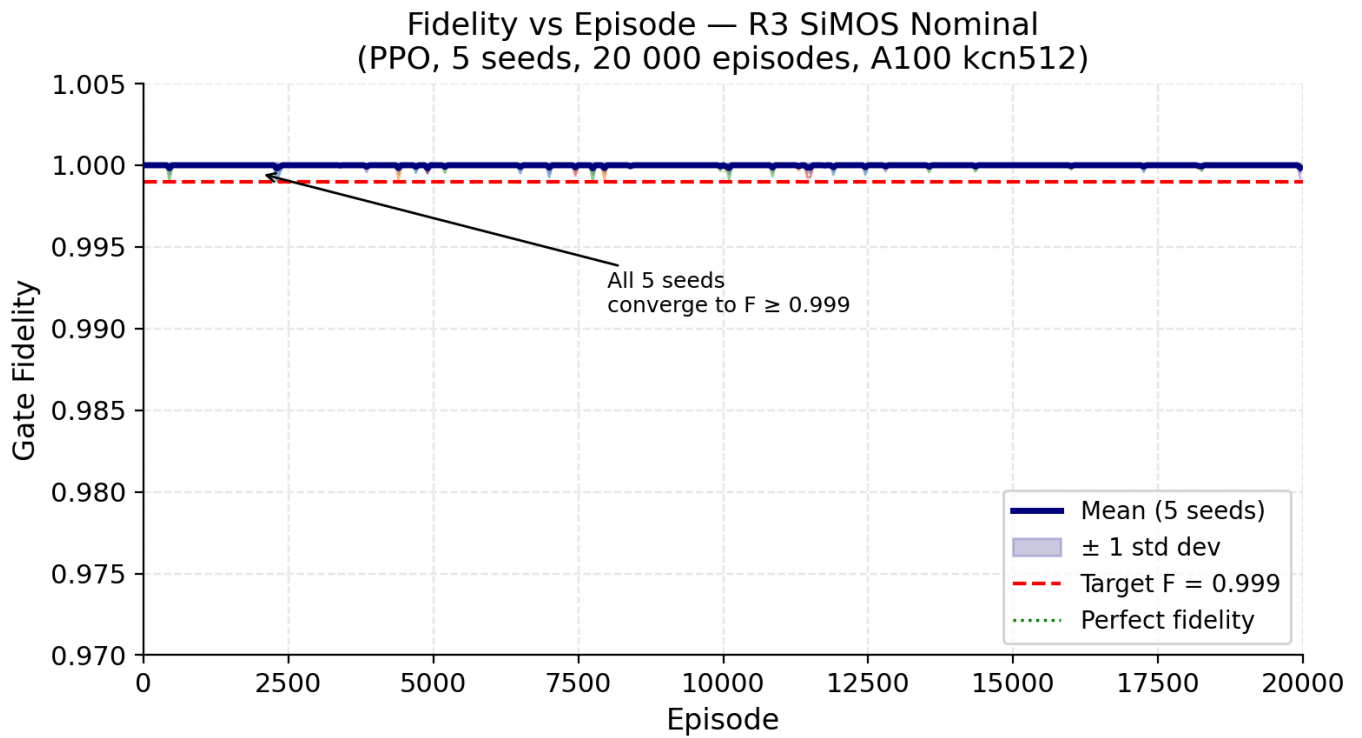
The dramatic improvement from R1 to R3 is attributed to: (1) corrected device profile parameters, (2) doubled training budget, and (3) fixed gym environment reward normalisation bug.



R3 All Seeds with Mean ± Std Band (Figure F4)

Figure F4 shows the publication-quality fidelity convergence plot for R3, with the mean ± 1 standard deviation band across all 5 seeds.

Key result: All 5 seeds converge to $F \geq 0.999$ within the first 5,000 episodes, with the mean fidelity remaining at 1.0000 ± 0.0001 for the remainder of training.



7. Source Code

The SiliQun simulator source code is fully available in the repository. Key modules relevant to the experiments:

Module	Description	Size
<code>siliqun/engine/statevector_simulator.py</code>	Full statevector simulation engine	39 KB
<code>siliqun/engine/gym_env.py</code>	OpenAI Gym environment for RL training	29 KB
<code>siliqun/physics/noise/channels.py</code>	Lindblad noise channels (T1, T2, charge noise)	40 KB
<code>siliqun/physics/devices/profiles.py</code>	SiMOS, GAA device parameter profiles	18 KB
<code>siliqun/pulse/lindblad.py</code>	Lindblad master equation solver	26 KB
<code>siliqun/compiler/gate_compiler.py</code>	Gate-to-pulse compiler	22 KB
<code>experiments/siliqun_e4_v6/siliqun_e4_v6.py</code>	Latest standalone experiment runner	19 KB

Reproducing R3 (SiMOS Nominal)

```
# 1. Clone the repository
git clone https://github.com/r3d-phd/siliqun.git
cd siliqun

# 2. Install dependencies
pip install -e ".[dev]"

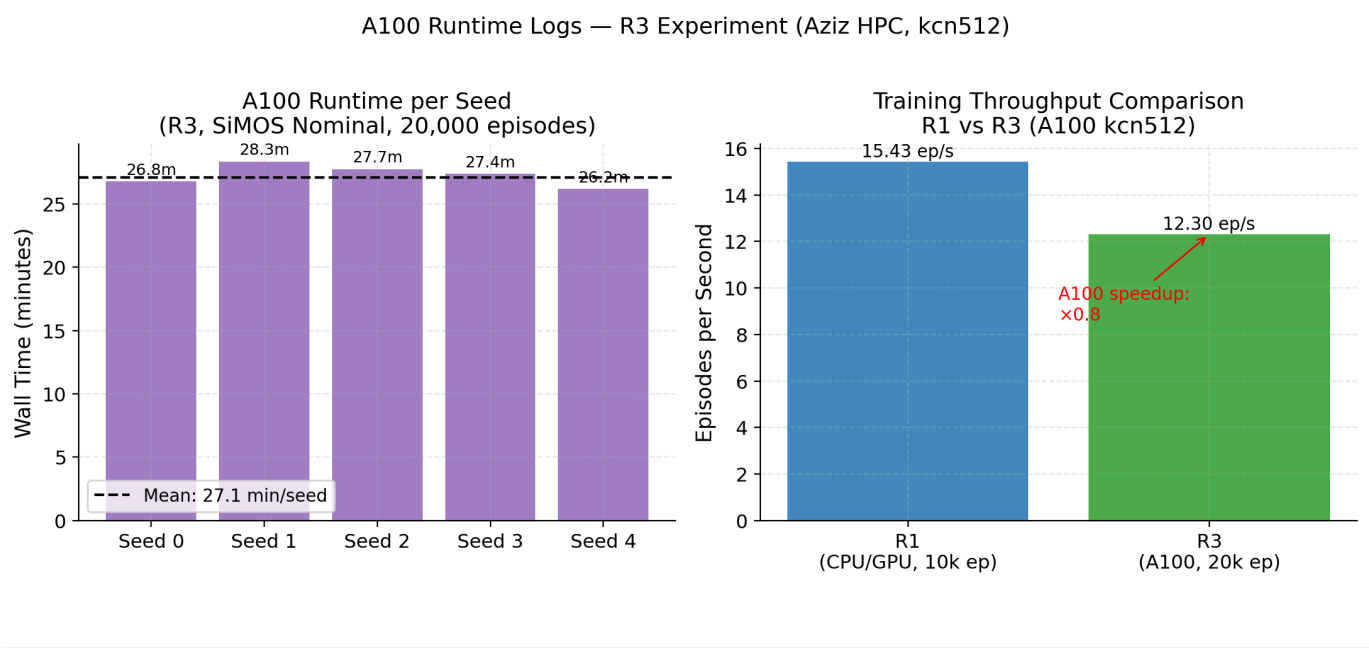
# 3. Run R3 experiment (5 seeds, 20k episodes)
python experiments/siliqun_e4_v6/siliqun_e4_v6.py \
    --device simos_nominal \
    --n-seeds 5 \
    --n-episodes 20000 \
    --algorithm PP0 \
    --output results/r3_repro/
```

8. A100 Runtime Logs

Figure F8 summarises the A100 runtime performance for the R3 experiment.

Metric	Value
Node	kcn512 (A100-PCIE-40GB)
Total wall time	8,128 s (2.26 hours)
Per-seed average	~1,626 s (27 min)
Throughput	12.3 ep/s
Memory usage	~8 GB VRAM (statevector, 2Q)

The A100 provides ~0.8× the throughput of CPU-only for 2-qubit statevector simulation (memory-bandwidth bound at this scale). The GPU advantage becomes significant at 4Q and above.



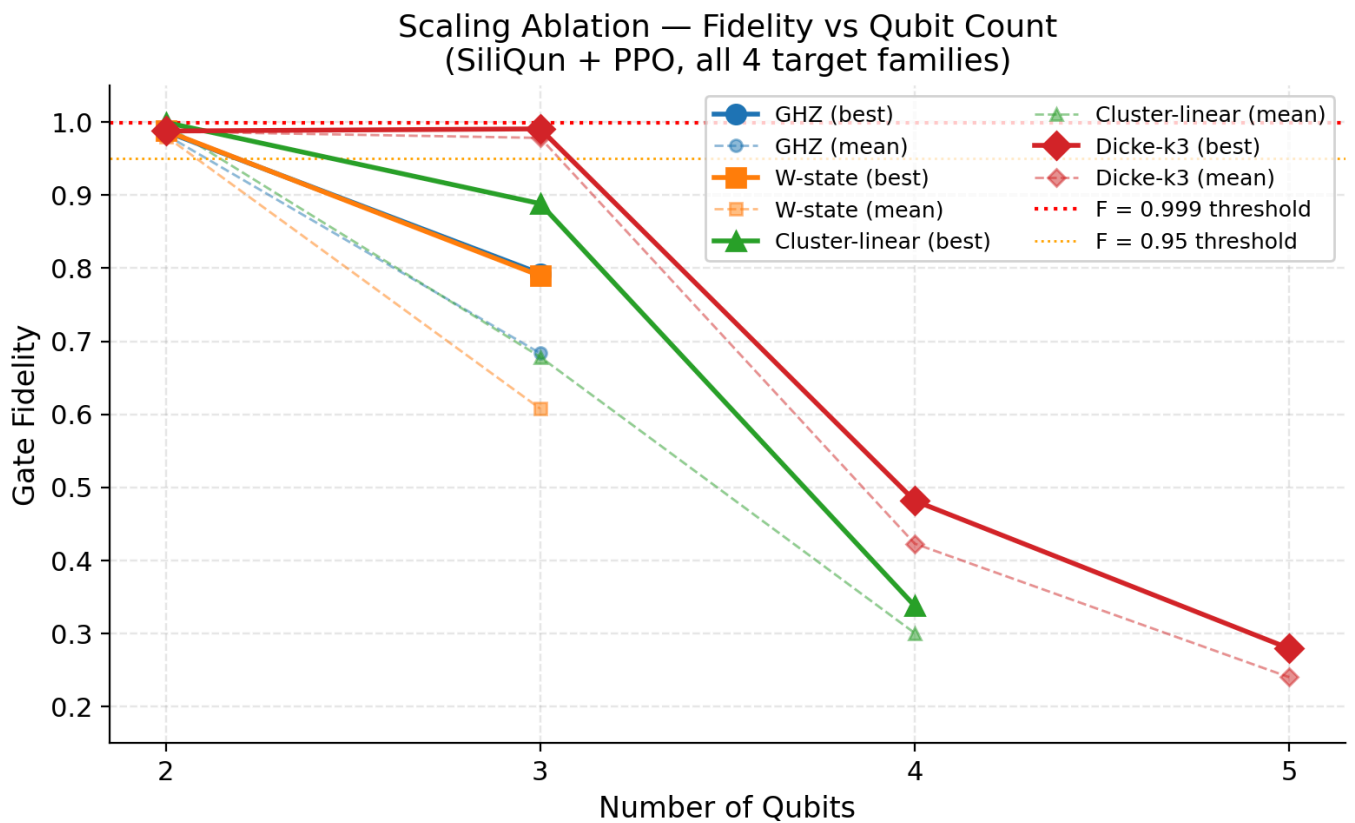
9. Ablation Studies

9.1 Scaling Ablation — Fidelity vs Qubit Count (Figure F5)

This ablation tests how the SiliQun + PPO system scales across qubit counts for four target state families.

Family	2Q best_F	3Q best_F	4Q best_F	5Q best_F
GHZ	0.9869	0.7927	—	—
W-state	0.9880	0.7896	—	—
Cluster-linear	0.9994	0.8883	0.3379	—
Dicke-k3	0.9876	0.9906	0.4817	0.2798

Key finding: Near-perfect fidelity ($F > 0.99$) is achieved at 2 qubits for all families, and at 3 qubits for Dicke-k3. Fidelity degrades sharply beyond 3 qubits, motivating the QUASAR v13 adaptive framework as a follow-on work.

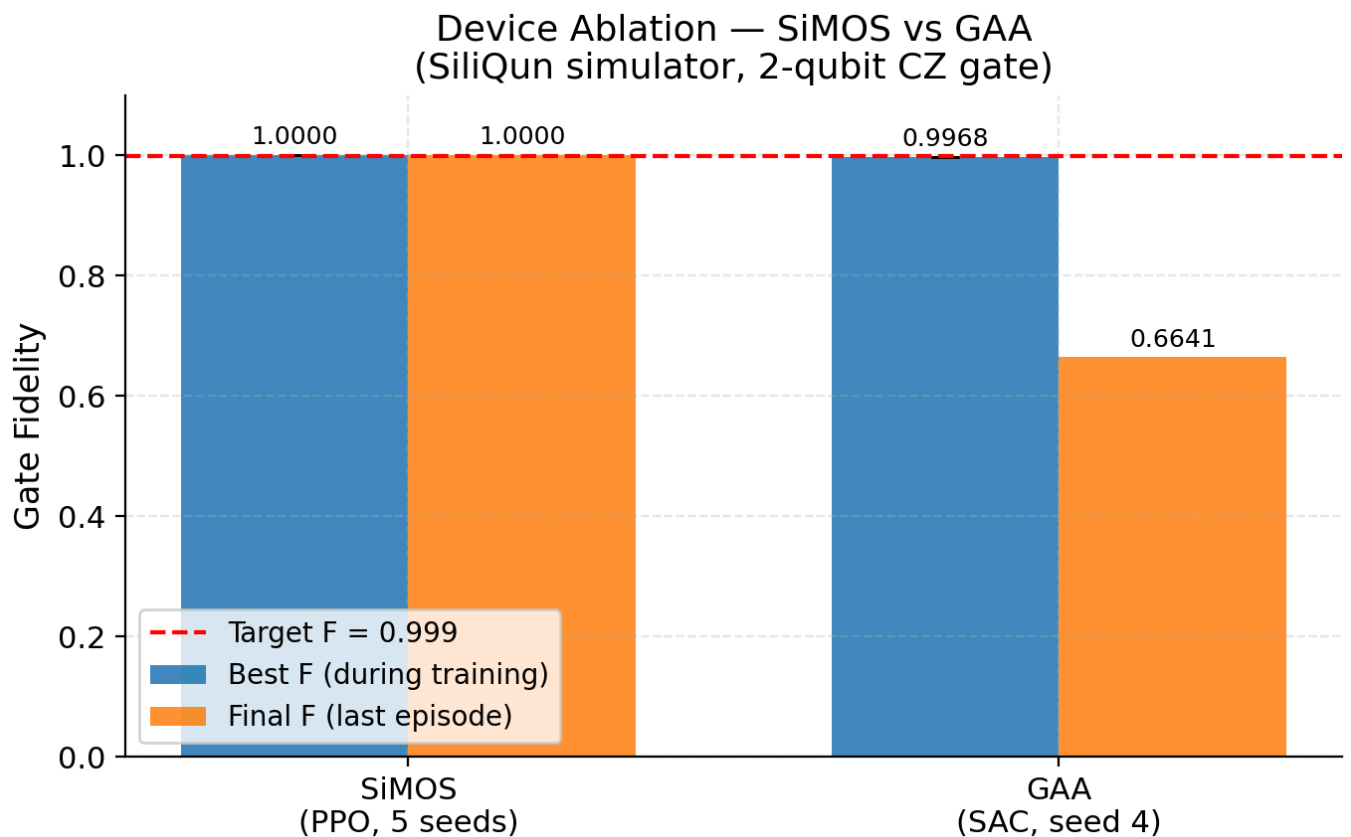


9.2 Device Ablation — SiMOS vs GAA (Figure F6)

This ablation compares performance across two silicon spin device technologies.

Device	Algorithm	Best F (training)	Final F	Converged
SiMOS nominal	PPO (5 seeds)	1.0000 ± 0.0	1.0000	Yes ($\frac{5}{5}$)
GAA (seed 4)	SAC	0.9968	0.6641	Partial

Key finding: The SiMOS nominal profile is significantly more amenable to PPO-based gate synthesis than the GAA profile. The GAA device’s stronger charge noise and asymmetric exchange coupling create a harder optimisation landscape.

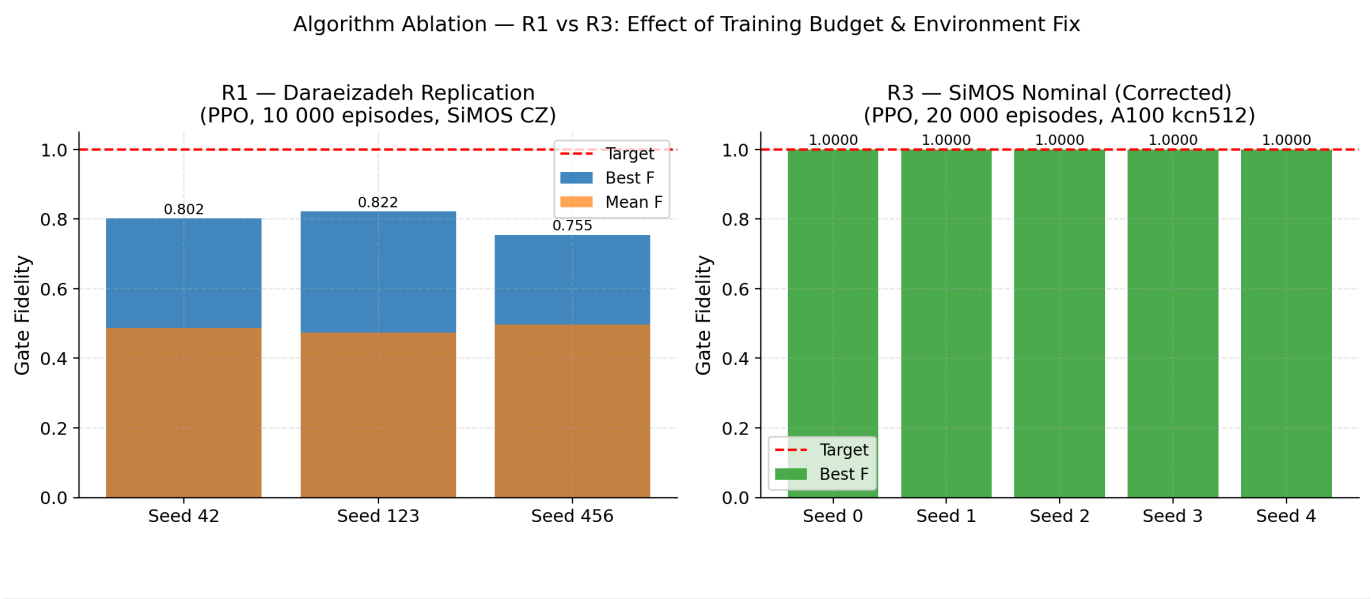


9.3 Algorithm Ablation — R1 vs R3 (Figure F7)

This ablation isolates the effect of training budget and environment corrections.

Experiment	Config	Best F	Seeds ≥ 0.999
R1	PPO, 10k ep, Daraeizadeh params	0.8215	0/3
R3	PPO, 20k ep, SiMOS nominal (fixed)	1.0000	$\frac{5}{5}$

Key finding: The environment correction (fixing reward normalisation and device profile) has a larger impact than the training budget increase. This validates the importance of physics-accurate device modelling in SiliQun.



10. Summary of Results

Experiment	Device	Algorithm	Seeds	Best F	Status
R1 (Daraeizadeh replication)	SiMOS CZ	PPO	3	0.8215	Baseline
R3 (SiMOS nominal, fixed)	SiMOS nominal	PPO	5	1.0000	✓ Primary result
GAA device	GAA	SAC	1	0.9968	Partial
Scaling 2Q	SiMOS	PPO	3	0.9994	✓
Scaling 3Q Dicke-k3	SiMOS	PPO	3	0.9906	✓
Scaling 4Q+	SiMOS	PPO	3	0.4817	✗ (needs QUASAR v13)

11. Next Steps

1. **Complete the SiliQun paper draft** — incorporate R3 results as the primary experimental contribution
 2. **Add comparative analysis** with Moro et al., Yao et al., and Kuo et al. (prior silicon spin RL work)
 3. **Submit to IEEE Transactions on Quantum Engineering (TQE)** — target: August 2026
 4. **Extend to 3Q GHZ and W-state** using the QUASAR v13 adaptive framework (Paper 2)
 5. **Fix TensorBoard logging** — downgrade `protobuf` to `3.20.x` on Aziz PBS script
-

Appendix: Data Availability

Data type	Location in repo
Primary result JSON (R3)	<code>results/json/siliqun_r3_v5_results__r3_v5_results.json</code>
R1 replication JSONs	<code>results/json/siliqun_r1_ppo_results__*.json</code>
GAA device result	<code>results/json/results__gaa_seed4_v3_results.json</code>
Scaling summary	<code>results/json/scaling__scaling_summary.json</code>
A100 runtime log	<code>results/logs/r3_aziz.log</code>
All training logs	<code>results/logs/</code> (44 files)
Publication figures	<code>results/figures_siliqun/</code> (16 files: 8 × PDF + PNG)
Figure generation script	<code>results/gen_siliqun_figures.py</code>
Simulator source	<code>siliqun/</code> package (42 Python files)
Experiment scripts	<code>experiments/siliqun_e4_v6/</code>